# Verifying DHCP

Difficulty (out of 5 stars): ★★★ Recommended study time: 2 hours

## Contents

# 1 About This Lab

## 1.1 Task Description

Generally, users expect that each terminal on a network can dynamically obtain network parameters such as the IP address, DNS server IP address, routing information, and gateway information. Users do not need to manually configure network parameters such as the terminal IP address. In addition, some mobile terminals (such as mobile phones, iPads, and laptops) require plug-and-play, and users do not need to modify network parameters such as the terminal IP address each time. To meet these requirements, the Dynamic Host Configuration Protocol (DHCP) server function can be configured on a user gateway at the aggregation layer or a core device to assign network parameters such as IP addresses to terminals.

DHCP uses the client/server model to dynamically configure and centrally manage network parameters for users. The DHCP server uses an address pool to assign network parameters such as IP addresses to users. There are two types of address pools: interface address pool and global address pool.

In this task, router R1 functions as the DHCP server. After completing this task, you will have a clear understanding of DHCP and can independently use DHCP to assign network parameters such as IP addresses to hosts.

## 1.2 Objectives

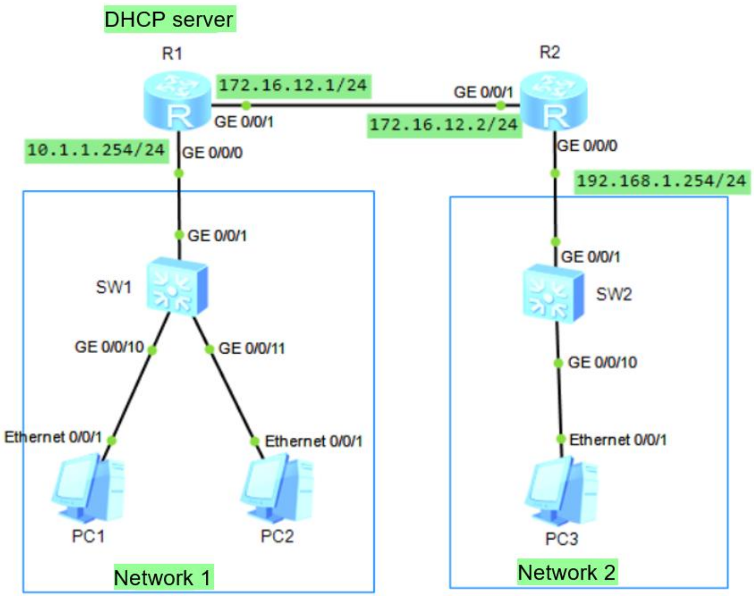
On completing this task, you will be able to:

1. Understand fundamentals of DHCP.
2. Be familiar with and master the DHCP server based on an interface address pool.
3. Be familiar with and master the DHCP server based on a global address pool.
4. Be familiar with and master the DHCP relay configuration.

# 2 Task Preparation

## 2.1 Network Topology

**Figure 1** DHCP lab topology




## 2.2 Initial Configuration

1. IP addresses
   - Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.1.1.254 24
[R1-GigabitEthernet0/0/0]quit
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 172.16.12.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

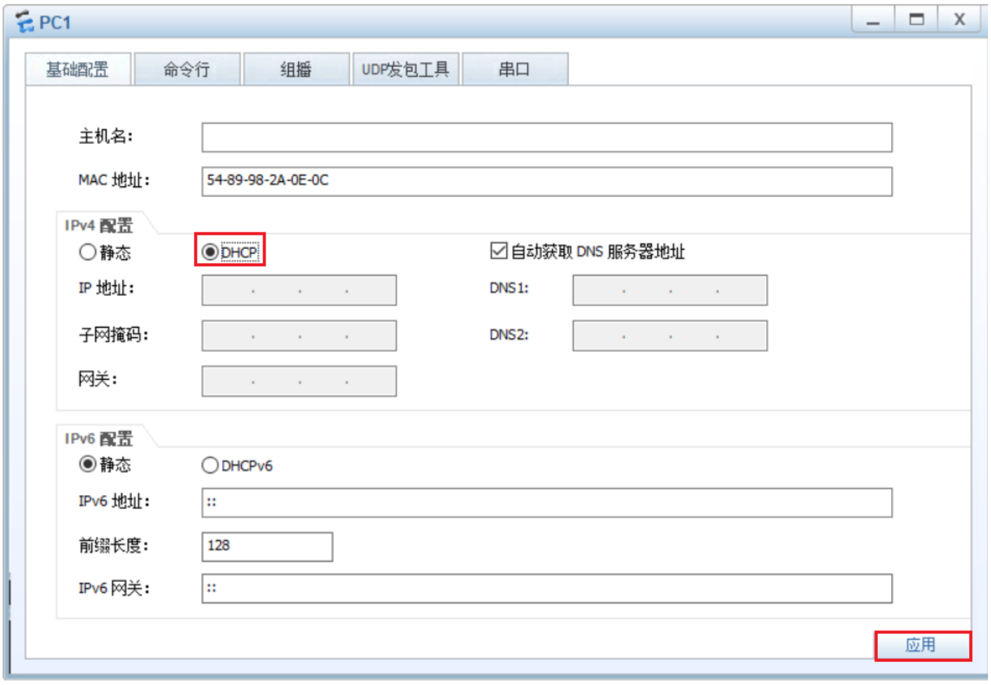
   - Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/1
```

```
[R2-GigabitEthernet0/0/1]ip address 172.16.12.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 192.168.1.254 24
[R2-GigabitEthernet0/0/0]quit
[R2]
```

- The initial configurations of SW1 and SW2 are not required. By default, all their interfaces are in VLAN 1.
- The IPv4 configurations of PC1, PC2, and PC3 are configured using DHCP. The following uses PC1 as an example to describe how to configure PC1 to obtain network parameters such as an IP address through DHCP. For details, see Figure 2.

**Figure 2** Configuring PC1 to obtain network parameters such as an IP address through DHCP




2. Route configuration
   - R1 route configuration

```
[R1]ip route-static 192.168.1.0 255.255.255.0 172.16.12.2
```

   - R2 route configuration

```
[R2]ip route-static 10.1.1.0 255.255.255.0 172.16.12.1
```

# 3 Task Implementation

## 3.1 Interface Address Pool

Configure R1 as a DHCP server that uses an interface address pool to assign IP addresses to PC1 and PC2.

The configuration of an interface address pool is simple. The interface address pool applies only to the scenario where users and the DHCP server are on the same network segment and network parameters such as IP addresses can be assigned only to users on the corresponding interface. It is applicable to small networks where the number of devices is limited and the configuration and maintenance workload is controllable. After the DHCP server function based on an interface address pool is configured on the user gateway, fixed hosts and mobile terminals connected to the corresponding interface can automatically obtain network parameters such as IP addresses, without manual configuration or modification.

**Step 1** Enable the DHCP service on R1. By default, the DHCP service is disabled.

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R1]
```

**Step 2** Configure an interface address pool. Configure an interface address pool on G0/0/0 of R1 to dynamically assign IP addresses to PC1 and PC2 connected to the interface.

- The **dhcp select interface** command enables the DHCP server function based on the interface address pool on an interface. By default, this function is disabled.
- The **dhcp server lease day 2** command sets the lease to two days. By default, the lease is one day.

```
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]dhcp select interface
[R1-GigabitEthernet0/0/0]dhcp server lease day 2
```

**Step 3** Enable the device to save DHCP data to the storage medium. If a fault occurs on the device, you can run the **dhcp server database recover** command after the system restarts to restore DHCP data from files on the storage medium.

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]dhcp server database recover
```

**Step 4** Configure terminals to automatically obtain IP addresses. On the IPv4 configuration pages of PC1 and PC2, select DHCP again and click **Apply** to enable the PCs to send DHCP requests again. Check the IP addresses of PC1 and PC2, as shown in Figure 3 and Figure 4.

**Figure 3** IP address obtained by PC1 through DHCP

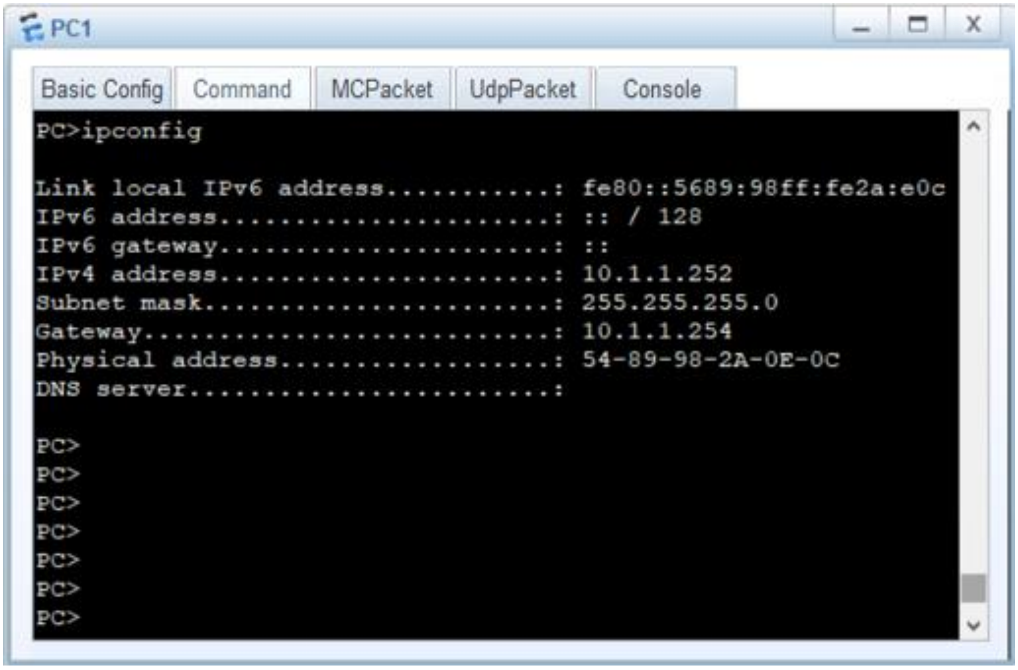
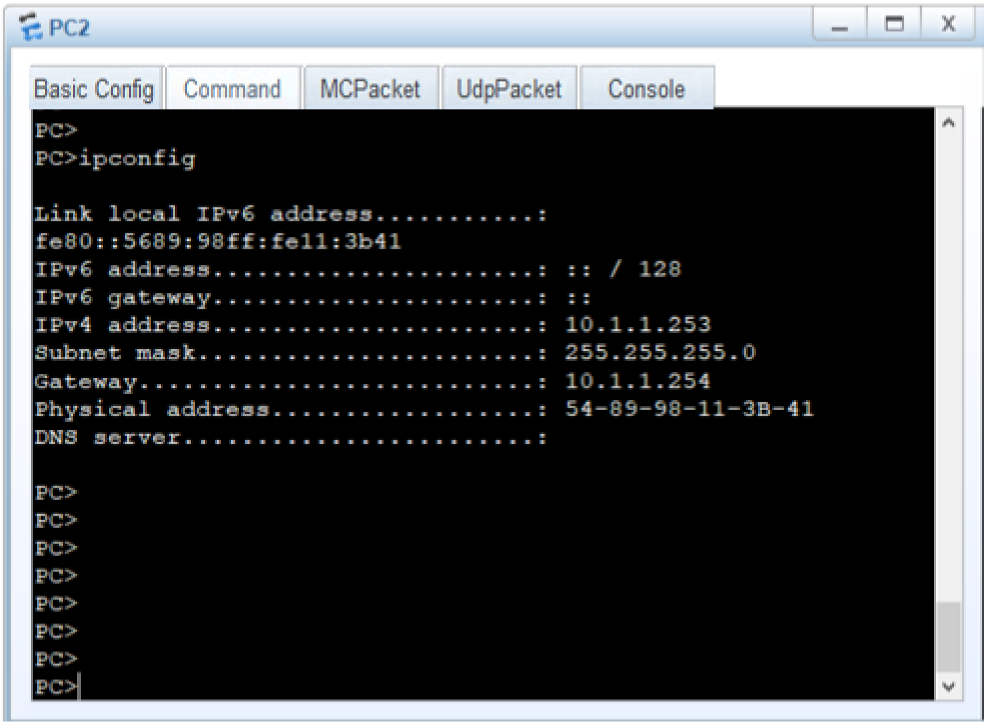
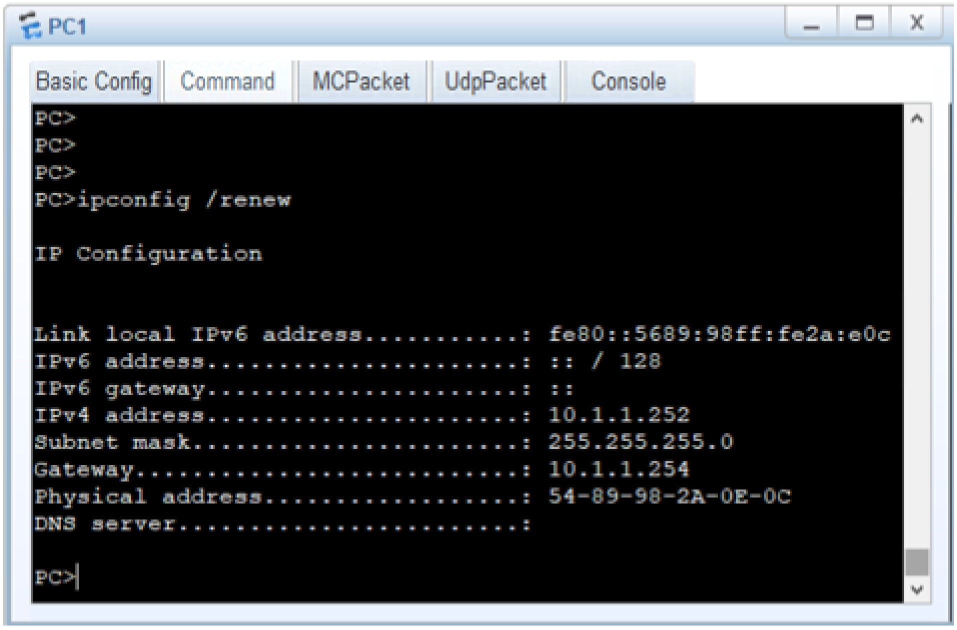



**Figure 4** IP address obtained by PC2 through DHCP




**Step 5** Run the **ipconfig /renew** command on a PC to apply for an IP address again, as shown in Figure 5.

**Figure 5** Applying for an IP address again




**Step 6** Check DHCP packets, as shown in Figure 6, Figure 7, Figure 8, and Figure 9.

**Figure 6** DHCP Discovery

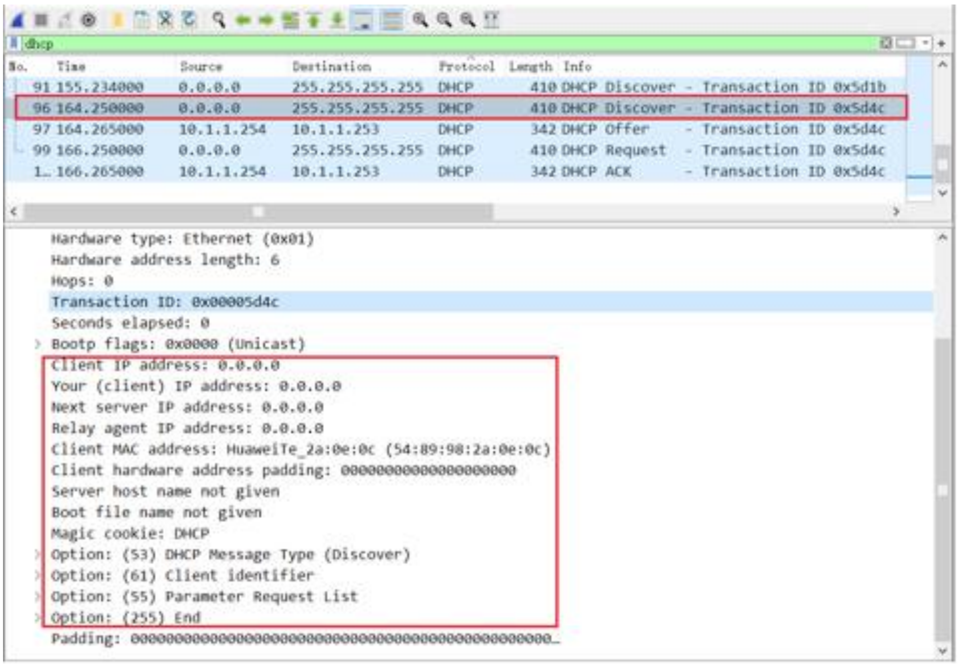


**Figure 7** DHCP Offer

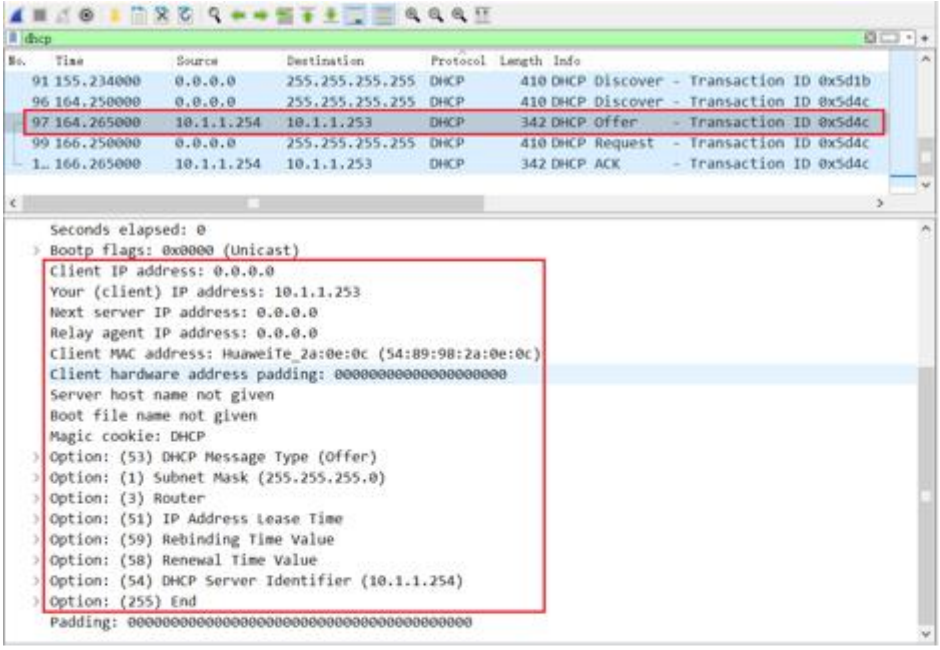



**Figure 8** DHCP Request

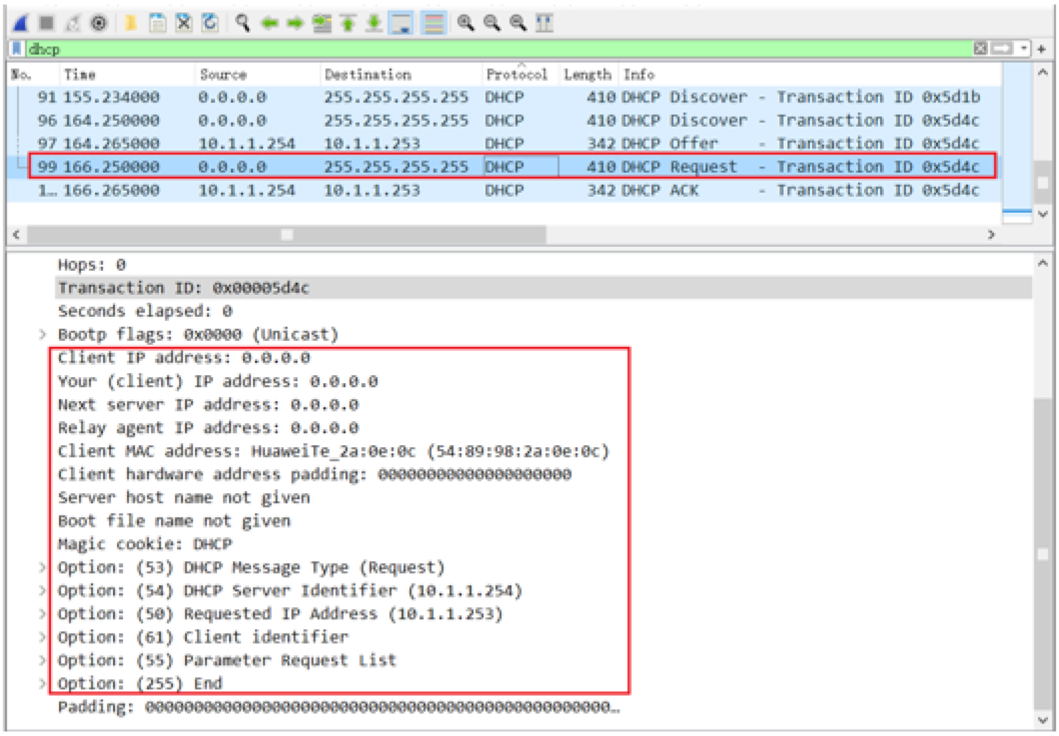
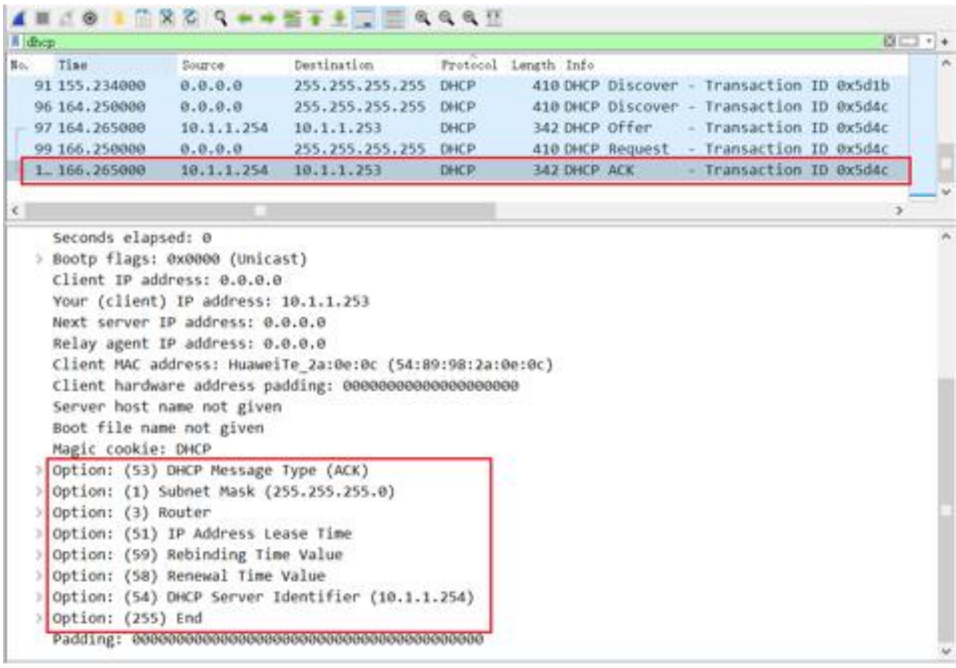


**Figure 9** DHCP ACK




## 3.2 Global Address Pool and Relay Agent

Compared with an interface address pool, a global address pool can be applied to large networks. It is recommended that the DHCP server function based on a global address pool be configured on a core device or a dedicated DHCP server be deployed in the server area to uniformly assign network parameters such as IP addresses. Only the DHCP relay function needs to be enabled on the user gateway.

PC3 is located on the network segment 192.168.1.0/24 of R2. When PC3 requests an IP address from the DHCP server R1, the DHCP request cannot be sent to the DHCP server R1 across Layer 3. Therefore, you need to configure the DHCP relay function on the interface connecting R2 to the network segment of PC3.

A DHCP relay agent forwards DHCP packets between the DHCP server and clients. When the DHCP server and clients are on different network segments, a DHCP relay agent needs to be configured. For DHCP clients, the DHCP relay agent is the DHCP server; for the DHCP server, the DHCP relay agent is a DHCP client.

In addition, an interface address pool cannot be used in this scenario, and a global address pool is required. This is because the configuration of an interface address pool is simple, and the interface address pool can be used only when users and the DHCP server are on the same network segment and network parameters such as IP addresses can be assigned only to users on the corresponding interface.

**Step 1** Configure a global address pool on R1. The address pool must be configured on the network segment where PC3 resides, the IP address of the interface connecting R2 to PC3 must be configured as the gateway address, the DHCP lease must be set to 10 days, and the DNS server address must be set to 114.114.114.114.

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
```

```
[R1]ip pool pool1
[R1-ip-pool-pool1]network 192.168.1.0 mask 255.255.255.0
[R1-ip-pool-pool1]gateway-list 192.168.1.254
[R1-ip-pool-pool1]lease day 10
[R1-ip-pool-pool1]dns-list 114.114.114.114
[R1-ip-pool-pool1]quit
[R1]
```

**Step 2** Run the **dhcp select global** command on the interface connecting R1 to R2, indicating that IP addresses need to be assigned to hosts from the global address pool.

```
<R1>system-view
Enter system view, return user view with Ctrl+Z.
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1] dhcp select global
[R1-GigabitEthernet0/0/1]quit
[R1]
```

**Step 3** Enable DHCP on R2.

```
<R2>system-view
Enter system view, return user view with Ctrl+Z.
[R2]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
```

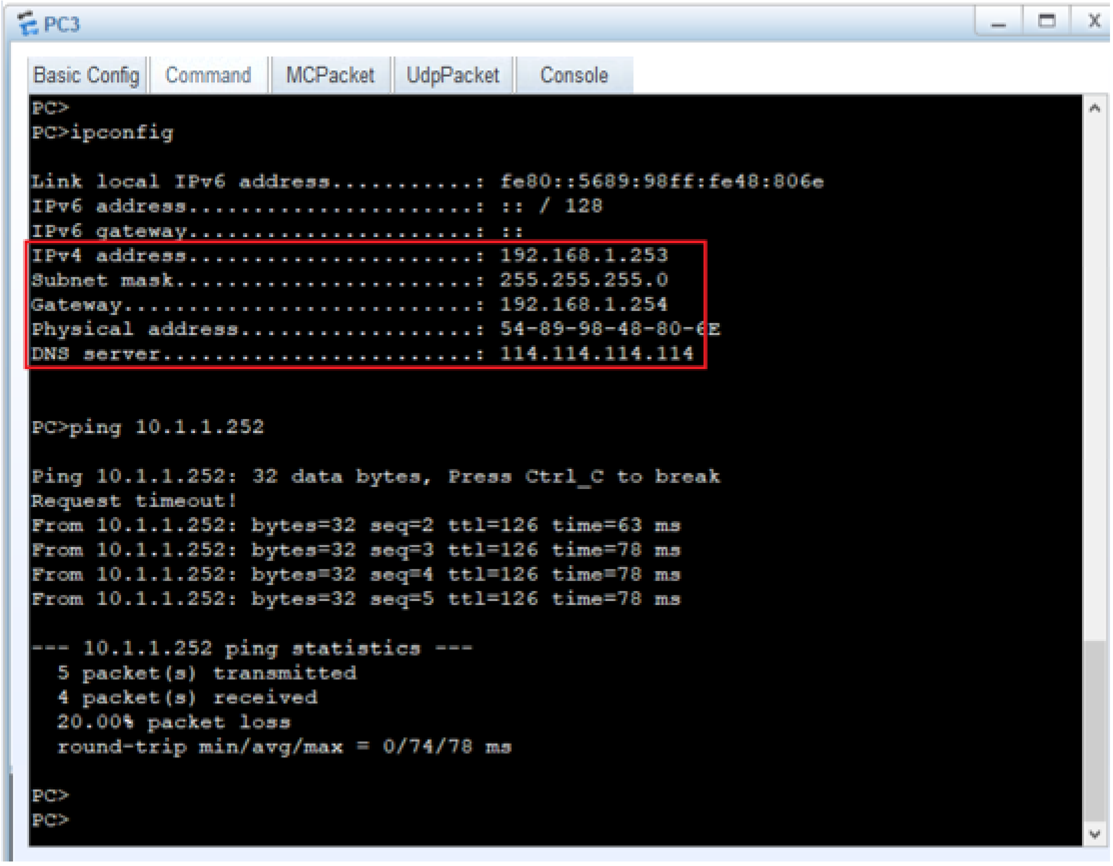
**Step 4** Configure the DHCP relay function on R2.

On R2, configure the DHCP relay function on G0/0/0 connected to PC3 to forward the DHCP request packets received from the gateway of PC3 to the DHCP server R1.

```
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]dhcp select relay
[R2-GigabitEthernet0/0/0]dhcp relay server-ip 172.16.12.1
[R2-GigabitEthernet0/0/0]
```

**Step 5** Configure terminals to automatically obtain IP addresses. On the IPv4 configuration page of PC3, select DHCP again and click **Apply** to enable the PC to send a DHCP request again. Check the IP address of PC3, as shown in Figure 10.

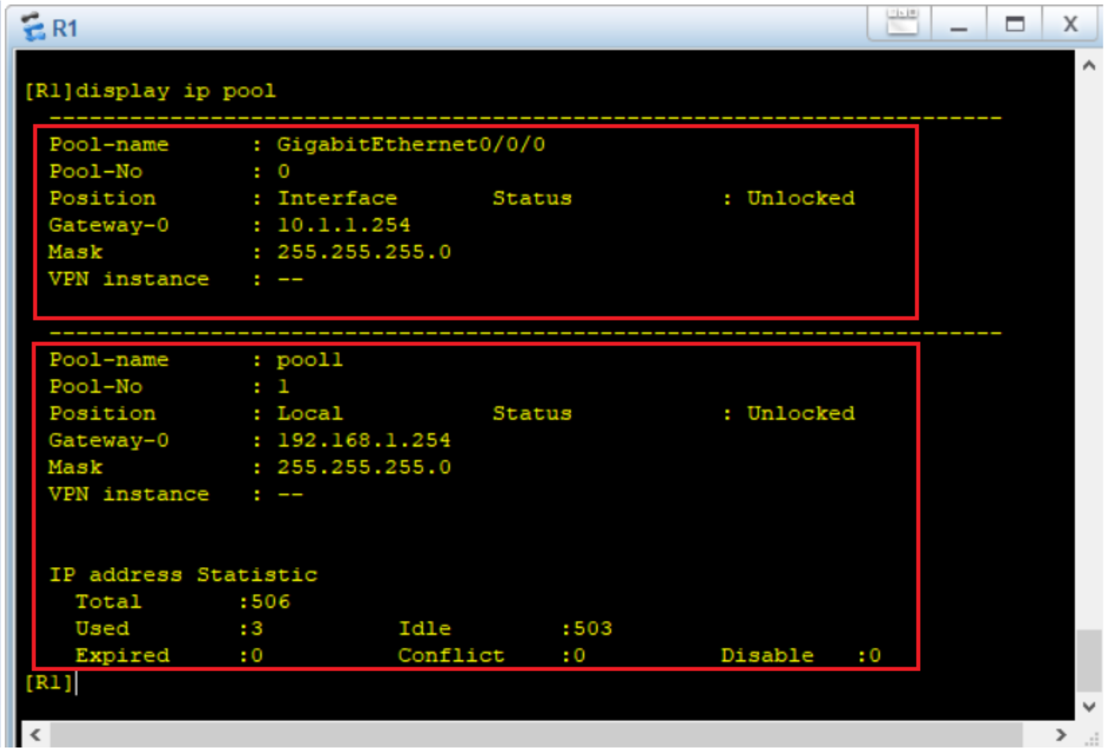
**Figure 10** IP address obtained by PC3 through DHCP




# 4 Verification

1. Check the DHCP address pool on R1. The command output shows an interface address pool and a global address pool, as shown in Figure 11.

**Figure 11** DHCP address pools on R1



```
R1
[R1]display ip pool
 -----------------------------------------------------------------------
  Pool-name       : GigabitEthernet0/0/0
  Pool-No         : 0
  Position        : Interface       Status            : Unlocked
  Gateway-0       : 10.1.1.254
  Mask            : 255.255.255.0
  VPN instance    : --

 -----------------------------------------------------------------------
  Pool-name       : pool1
  Pool-No         : 1
  Position        : Local           Status            : Unlocked
  Gateway-0       : 192.168.1.254
  Mask            : 255.255.255.0
  VPN instance    : --


  IP address Statistic
    Total       :506
    Used        :3          Idle        :503
    Expired     :0          Conflict    :0            Disable    :0
[R1]
```

2. Check the usage of the global address pool.

```
[R1]display ip pool name pool1
  Pool-name        : pool1
  Pool-No          : 1
  Lease            : 10 Days 0 Hours 0 Minutes
  Domain-name      : -
  DNS-server0      : 114.114.114.114
  NBNS-server0     : -
  Netbios-type     : -
  Position         : Local              Status              : Unlocked
  Gateway-0        : 192.168.1.254
  Mask             : 255.255.255.0
  VPN instance     : --
 -----------------------------------------------------------------------------
          Start              End        Total   Used   Idle(Expired)   Conflict   Disable
 -----------------------------------------------------------------------------
     192.168.1.1     192.168.1.254     253      1         252(0)           0          0
 -----------------------------------------------------------------------------
[R1]
```

# 5 Quiz

1. What are the application scenarios of an interface address pool?
2. What are the application scenarios of a global address pool?
3. Describe the DHCP relay function and its application scenarios.